

Ibrahim Shoukry

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EDUCATION

University of California, Berkeley

BS, Electrical Engineering & Computer Sciences

Berkeley, CA

Graduated December 2025

Coursework *Algorithms, Advanced Programming (C/C++), Internet Architecture & Protocols, Database Systems (Distributed Systems & DBMS Design), Computer Security (Cryptography, Networks), Artificial Intelligence*

TECHNICAL SKILLS

Languages: Python, Java, Go, TypeScript

Backend & Systems: REST APIs, JWT Authentication, Rate Limiting, Distributed Systems

Frameworks & Tools: React, Next.js, Git, Docker, Google Cloud Platform

Databases: PostgreSQL, MongoDB

EXPERIENCE

Software Engineer Intern

May 2025 – July 2025

Moccet

San Francisco, CA

- **End-to-End Ownership:** Led the design and implementation of the client onboarding lifecycle, integrating payment gateways and automated contract generation.
- **AI & Automation:** Developed an LLM-driven agentic workflow that automates project scoping, parsing unstructured client data into structured JSON requirements to remove manual data entry bottlenecks.
- **Migration & Modernization:** Migrated the production monolith to **Next.js** and **TypeScript**, establishing a strongly-typed codebase that significantly reduced runtime errors and improved developer velocity.
- Collaborated directly with founders to define the technical roadmap, refactoring legacy components to minimize technical debt while maintaining a rapid release cycle.

Software Engineer Intern

May 2024 – Aug 2024

Evitado Technologies

Hamburg, Germany

- Developed a **LiDAR-based system** to detect and segment moving objects near commercial aircraft, enhancing situational awareness for on-ground risk assessment.
- Researched and evaluated a range of **deep learning models** to determine the optimal solution for real-time object segmentation and motion prediction.
- Enhanced an open-source LiDAR processing pipeline by integrating support for custom geospatial data formats (including **rosbags**), ensuring seamless compatibility with proprietary datasets.
- Collaborated with cross-functional teams by presenting technical findings and proposing process improvements, which contributed to refined deployment strategies.

PROJECTS

Agent Authenticator | *Golang, JWT (RS256), Rate Limiting, Distributed Systems*

[Github Repo](#)

- **Production-Grade Auth:** Built a centralized identity gateway implementing **JWT (RS256)** signing, expiration strategies, and RSA key rotation to ensure secure service-to-service communication.
- **Distributed Rate Limiting:** Engineered a concurrency-safe **token-bucket algorithm** in Go, utilizing mutexes to manage per-tenant quotas and prevent resource exhaustion under load.
- **Middleware Architecture:** Developed reusable Go middleware for request logging, panic recovery, and trace context propagation, standardizing observability across services.

End-to-End File Sharing System | *Go, Cryptography, (RSA/Argon2)*

- Implemented a secure storage system using **hybrid encryption**: AES-GCM for file data and **RSA-2048** for key exchange, ensuring strict data confidentiality.
- Utilized **Argon2** for key derivation and **HMAC-SHA256** for message authentication, protecting against rainbow table attacks and data tampering.
- Designed a threat model explicitly handling revoked users and malicious server administrators, ensuring forward secrecy and tamper resistance.